Seokwon Jung

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Research Interests

I work on how LLM agents find, use, and build on knowledge: **information retrieval and RAG** beyond semantic similarity, **self-evolving agents** that update their memory and knowledge from experience, and **research automation** with LLMs for scientific discovery. I approach these problems from the data side as much as the method side, building the evaluation and training data that existing work lacks alongside methods that hold up on it.

Education

KAIST, M.S. in Artificial Intelligence — Mar 2026 – Feb 2028 (expected)
Kim Jaechul Graduate School of AI; advised by Prof. Seong Joon Oh

KAIST, B.S. in Computer Science and Chemistry (double major) — Mar 2014 – Feb 2026

Publications

[1] **MEME: Multi-Entity & Evolving Memory Evaluation.**
Seokwon Jung, Alexander Rubinstein, Arnas Uselis, Sangdoo Yun, Seong Joon Oh.
NeurIPS 2026, Evaluations & Datasets Track. [pdf] [code]

[2] **Cross-Pollination Retrieval: Finding Transferable Ideas Across Fields in Scientific Literature.**
Seokwon Jung, Yerang Kim, Yejin Kim, Alexander Rubinstein, Arnas Uselis, Sangwoo Park, Seong Joon Oh, Sangdoo Yun.
Preprint, 2026. [pdf]

[3] **TULIP: Targeted LLM Unlearning at Layers Identified Per-Input.**
Yejin Kim, William F. Shen, **Seokwon Jung**, Daeun Park, Seong Joon Oh.
Preprint, 2026. [pdf]

[4] **Form Over Content in Gradient-Based Data Attribution Methods.**
Sunwoo Kim, **Seokwon Jung**, SoHyung Kim, Seong Joon Oh, Alice Oh.
NeurIPS 2026 Workshop on Attributing Model Behavior at Scale (ATTRIB). [pdf]

[5] **Break the Output Geometry for Large Language Model Unlearning.**
Yejin Kim, William F. Shen, **Seokwon Jung**, Seong Joon Oh.
ICML 2026 Workshop on the Impact of Memorization on Trustworthy Foundation Models. [pdf]

[6] **Polyphonic Sound Event Detection Using Convolutional Bidirectional LSTM and Synthetic Data-based Transfer Learning.**
Seokwon Jung, Jungbae Park, Sang Wan Lee.
ICASSP 2019, pp. 885–889. (41 citations) [pdf]

Research Experience

Bluesignum — Research Scientist — Jul 2024 – Aug 2024

- Set the **research direction and system design** for emotional speech synthesis and for coupling a conversational model to a speech pipeline

Humelo Inc. — Co-founder, Research Scientist — Apr 2018 – Apr 2024

- Research on **controllable speech synthesis**, taken from model to deployment

- Reached MOS 3.5+ by conditioning generation on emotion embeddings with an Arousal–Valence–Dominance formulation
- Extended a Neural Source Filter vocoder to multiple speakers via speaker embeddings and added pitch conditioning from a neural pitch tracker

- Research on **synthetic-data transfer learning** for polyphonic sound event detection [3]
  - Improved F1 by 28.4% over the prior architecture on TUT 2016
- Research on the **training corpus** that made expressive synthesis learnable
  - Defined 9 emotion categories over 40+ hours of speech, with an annotation protocol for emotional state and noise
  - Reused my sound event detection model as an automatic quality filter to strip noisy segments from the training set

Uneedcomms — Research Intern Oct 2017 – Feb 2018

- Built the **data collection pipeline** that made ML feasible, as no training data had previously been collected
  - Designed the schema and query layer from scratch, then automated performance reporting over it (Presto, Hadoop, Spark, Flask)

Patent

"Method for Training a Sound Event Detection Model" 2019

KR 10-2019-0007575; PCT/KR2019/012932.

Academic Service

Co-organizer — **PALM: Personalized, Aligned, Long-term Memory for AI Systems**, NeurIPS 2026 Workshop.

Skills

- **Languages:** Python, SQL, C++, Java
- **ML:** PyTorch, Hugging Face, DSPy, scikit-learn; LLM fine-tuning and evaluation, transfer learning, model interpretability
- **Retrieval:** sparse and dense retrieval (BM25, embeddings), cross-encoder reranking, RAG, vector stores (Qdrant), knowledge graphs (Neo4j)
- **Data:** Spark, Hadoop, Presto, Docker; synthetic data generation, LLM-as-judge pipelines with human calibration, A/B-style controlled ablations